# Adaptive Negotiation Agent with Real-Time Strategy Detection

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## Abstract

This study proposes a multi-agent large language model (LLM) negotiator system that improves outcomes by explicitly modeling an opponent's negotiation strategy during dialogue. This multi-agent adaptive negotiation system has a dedicated *Profiler* agent that analyzes the opponent's language, infers a latent negotiation style, and provides a counter-strategy to a separate *Actor* agent responsible for generating the next response. The study benchmarks the adaptive negotiation system on NegotiationArena, with seller–buyer bargaining across multiple price scenarios and opponent personas, comparing with a baseline single-agent negotiator, and a stronger same-model comparison setup. The results show that explicit opponent modeling improves expected surplus in both seller and buyer roles, with a trade-off at the cost of lower deal rate and longer negotiations. The study further reveals that prompt-level assertiveness and aggressive initial anchoring have a large effect on outcomes across all systems, revealing a systematic exploit of the benchmark that complicates interpretation of payoff gains. The study further extend the setting to dynamic opponents that switch strategies during negotiation. The result shows that the profiler architecture remains competitive, though strategy-identification accuracy is limited in short interactions. These findings suggest that separating opponent modeling from surface-level response generation is a promising direction for negotiation agents, while also highlighting the need for more robust evaluation environments and stronger controls for prompt-induced reward hacking.

## 1 Introduction

Negotiation is a central capability for autonomous agents that interact with one another over limited resources, prices, or conflicting preferences. In realistic bargaining, an effective negotiator must do more than respond fluently: it must infer the opponent's likely strategy, adapt its own behavior over time, and balance cooperative language with competitive optimization. However, current LLM-based negotiation agents often react directly to the latest message without explicitly modeling the opponent, making them vulnerable to adversarial styles and prone to overly accommodating behavior.

This paper studies whether a *multi-agent decomposition* can improve negotiation performance. Instead of relying on a single LLM to both interpret the opponent and generate a response, we separate these functions into two modules: a *Profiler*, which analyzes the dialogue and predicts the opponent's negotiation style, and an *Actor*, which receives the profiler's recommendation and produces the next utterance. Our central question is: *can a multi-agent negotiator detect opponent strategies and adapt in ways that improve negotiation outcomes?*

We implement this approach in a seller–buyer bargaining environment built on NegotiationArena. The profiler reasons over a small set of opponent personas—including hardball, sycophant, stalling,

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friendly, and neutral—and proposes next step strategies with counter-tactics. We compare three settings: a baseline single-agent negotiator, our profiler-based negotiator, and a same-model comparison system intended as a stronger reference point. Across multiple scenarios, we evaluate deal rate, expected surplus, average turns, and persona-conditioned behavior.

Our contributions are threefold. First, we propose a simple but effective decomposition of negotiation into *opponent modeling* and *response generation*. Second, we provide an empirical study of this architecture across both seller and buyer roles, showing that explicit profiling improves expected surplus but may reduce deal rate and increase negotiation length. Third, we analyze a benchmark artifact—namely that aggressive initial anchoring can systematically inflate payoffs regardless of model—and discuss its implications for evaluating negotiation agents. We also present an extended study with dynamic opponents that switch strategies mid-dialogue.

## 2 Related Work

### 2.1 LLM Negotiation Frameworks

A LLM negotiation framework is a system that enables AI agents powered by LLMs to communicate, reason, and reach agreements through structured dialogue protocols and decision strategies [Li et al., 2025, 2023, 2024]. This study is benchmarked on NegotiationArena [Bianchi et al., 2024], which evaluates LLMs in multi-turn negotiation across resource exchange, ultimatum, and sellerbuyer games. It showed that behavioral personas ("cunning," "desperate") improve GPT-4's payoff by up to 20%, but tested only two hand-crafted personas against a default opponent with no adaptive counter-strategy. [Abdelnabi et al., 2024] complements this with multi-party, multi-issue scorable games, showing LLM agents are vulnerable to adversarial incentive structures. Most closely related is ASTRA [Kwon et al., 2025], which performs turn-level offer optimization via opponent preference inference, LP-based counteroffer generation, and acceptance probability estimation. ASTRA improves negotiation outcomes across diverse opponent types in multi-issue bargaining but operates entirely within a single agent pipeline. Standard corpora such as CaSiNo [Chawla et al., 2021] and CraigslistBargains [He et al., 2018] underpin much of this work.

### 2.2 Opponent Modeling in LLM Agents

Opponent modeling in LLM agents studies how large language model-based agents infer, predict, and adapt to other agents' beliefs, preferences, and strategies during interaction to enable more effective strategic decision-making [Kuru et al., 2025, Xia et al., 2024, Akata et al., 2025, Lorè and Heydari, 2024]. EMO [Yu et al., 2025] constructs per-opponent models refined via bi-level self-refinement and global validation, significantly improving decision-making in deduction games (Avalon, Who is the Undercover)—but it models hidden game state, not behavioral style, and feeds into a single reasoning module. Suspicion-Agent [Guo et al., 2023] decomposes card game play into sub-modules (observation interpreter, planning module with theory-of-mind-aware reasoning), achieving competitive performance against classical algorithms such as CFR—but as a single agent on discrete action spaces, not natural language negotiation. Across these works, opponent modeling remains embedded within one agent's pipeline.

### 2.3 Decoupling Strategy from Execution

Decoupling strategy from execution studies how separating high-level strategic planning from action generation can improve the controllability and robustness of LLM agents [Yao et al., 2023, Shinn et al., 2023, Hu et al., 2025]. He et al. [2018] proposed separating negotiation into coarse dialogue act selection and retrieval-based generation, avoiding degenerate RL solutions while maintaining fluent output—but their strategic module selects dialogue acts, not opponent models. Du et al. [2024] showed that multi-agent debate improves factuality over single-agent generation, establishing that role decomposition across LLM agents is a viable design pattern. Our work synthesizes these lines: we extend NegotiationArena with diverse opponent personas and introduce a two-agent architecture where a dedicated profiler identifies the opponent's negotiation style and provides a counter-strategy to a separate, smaller speaker agent, investigating whether this decomposition improves a weaker negotiator's performance against diverse adversaries.

In contrast to prior work, our approach explicitly separates opponent modeling from surface-level response generation using two interacting agents. We evaluate whether this decomposition improves negotiation outcomes against diverse opponent personas.

## 3 Method

### 3.1 Negotiation Task

We consider a bilateral seller–buyer negotiation game. The seller has production cost $c$ and the buyer has willingness-to-pay $w$. If the agents agree on a final price $p$, the seller receives utility $p - c$ and the buyer receives utility $w - p$. The zone of possible agreement (ZOPA) is the interval of mutually beneficial prices, i.e.,

$$[c, w],$$

when $w > c$. If no agreement is reached, both agents receive utility $0$.

Although agreements outside the ZOPA are economically irrational under this payoff model, we retain them in our analysis because they reveal failure modes of the agents, including poor utility tracking, benchmark artifacts, and boundedly rational behavior.

### 3.2 Profiler–Actor Architecture

Our approach separates negotiation into two components: a **Profiler**, which performs opponent modeling, and an **Actor**, which generates the next negotiation utterance. At turn $t$, given dialogue history $H_t$, the profiler infers the opponent's likely negotiation strategy and produces a counter-strategy instruction for the actor. The actor then generates the next response conditioned on both the dialogue history and the profiler's recommendation. Formally,

$$s_t = \text{Profiler}(H_t), \qquad m_t = \text{Actor}(H_t, s_t),$$

where $s_t$ denotes the profiler's strategic assessment and $m_t$ the next message.

The profiler is prompted to produce three outputs: (1) observations about the opponent's language, (2) an internal strategy inference over a fixed set of candidate negotiation styles, and (3) an actionable recommendation for the actor. This recommendation may include tactics such as holding firm on price, imposing a deadline, mirroring cooperative language, or pressing for a concrete offer. The actor does not independently reason about opponent type; instead, it conditions on the profiler's output to generate the next negotiation move. Figure 1 illustrates the interaction between the two modules.

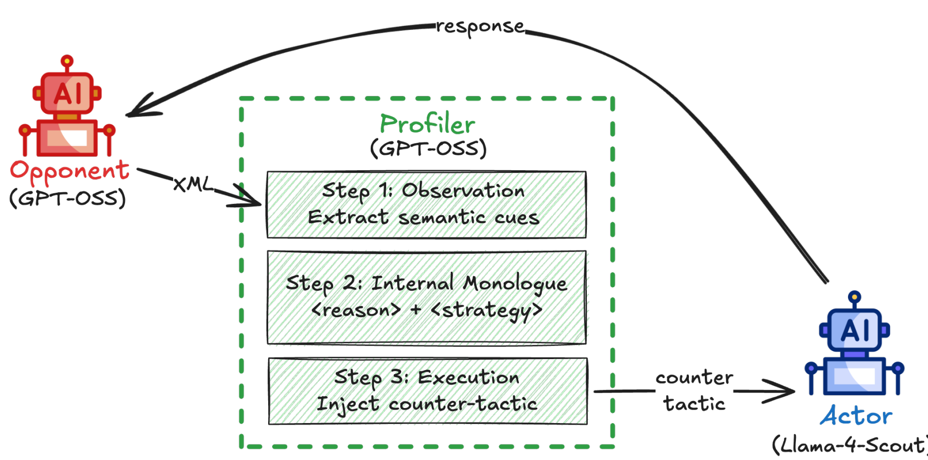



Figure 1: Profiler–Actor negotiation architecture. The profiler analyzes dialogue history to infer opponent strategy and outputs a counter-strategy used by the actor to generate the next response.

### 3.3 Opponent Strategy Space

We model opponents using five strategy types: **Neutral**, **Hardball**, **Friendly**, **Sycophant**, and **Stalling**. These categories are intended to capture recurring behavioral patterns in bargaining language, such as rigid pricing, excessive politeness, cooperative framing, or avoidance of concrete offers. The profiler maps the observed dialogue to one of these candidate strategies and selects a corresponding counter-tactic. Full pseudocode for the interaction loop is provided in Appendix B, and the full profiler prompt is included in Appendix A.

## 4 Experimental Setup

### 4.1 Task, Systems, and Evaluation Protocol

We evaluate our method in a bilateral seller–buyer bargaining game built on NegotiationArena. In each game, the seller has production cost $c$ and the buyer has willingness-to-pay $w$, so the zone of possible agreement (ZOPA) is $w - c$. We test 20 price scenarios spanning low-, mid-, and high-price regimes, including both positive-ZOPA settings and several negative-ZOPA settings where no mutually beneficial agreement exists. The full scenario list is provided in Appendix C (Table 1).

We compare three systems: (1) a **Baseline** single-agent negotiator using Llama-4-Scout against a `gpt-oss-120b` opponent; (2) our **Profiler** system, in which a Llama-4-Scout negotiator is guided by a separate `gpt-oss-120b` profiler; and (3) a stronger **Compare** reference in which the negotiator also uses `gpt-oss-120b`. Opponents are instantiated using five personas: *Neutral*, *Hardball*, *Friendly*, *Sycophant*, and *Stalling*. Each negotiation is capped at 10 turns, and we evaluate all three systems in both seller and buyer roles. This yields one evaluation for every scenario–persona pair under each system and role.

### 4.2 Metrics and Statistical Analysis

For a successful deal at price $p$, seller profit is $p - c$ and buyer surplus is $w - p$. We report the focal agent's normalized surplus,

$$\text{surplus\%} = \frac{\text{agent utility}}{w - c} \times 100,$$

for positive-ZOPA scenarios. Failed negotiations contribute 0 surplus. Our primary metric is therefore **expected surplus**, defined as

$$\mathbb{E}[\text{surplus}] = P(\text{deal}) \cdot \mathbb{E}[\text{surplus} \mid \text{deal}],$$

which combines value capture with deal frequency.

We also report **deal rate**, **average turns**, and the number of **outside-ZOPA deals**, i.e., agreements in which one party receives negative utility under the simulator's payoff model. Negative-ZOPA scenarios are analyzed separately, where the correct behavior is to reject the deal.

For system comparison, we report mean deltas in expected surplus together with bootstrap confidence intervals, and we use paired sign tests on matched scenario–persona runs to measure whether one system more often outperforms another.

## 5 Results

### 5.1 Main Findings

Across both seller and buyer settings, the profiler-based system improves expected surplus relative to the baseline. However, these gains are accompanied by two trade-offs: lower deal rate in some settings and longer negotiations on average.

### 5.2 Profiler as Seller

As the seller, the Profiler achieves a $51.20\%$ surplus improvement over the Baseline, while the Compare model (api-gpt-oss-120b) shows a smaller $25.00\%$ gain that is not statistically significant

(Figure 2). This suggests that the Profiler is more effective at capturing value within the ZOPA. Although the Profiler attains the highest average payoff, it has the lowest deal rate among the three systems ($87\%$) and takes the most turns on average ($4.4$) to conclude a negotiation (Figure 3). Rather than prioritizing rapid agreement, the Profiler exhibits strategic persistence, which can safeguard against aggresive or even malicious "low-ball" buyers.

| Condition | Mean Δ | CI Low (95%) | CI High (95%) | Std Dev | n | Sign-test p |
|---|---|---|---|---|---|---|
| Profiler vs Baseline | 51.20% | 29.0% | 75.8% | 110.5% | 85 | 0.06 |
| Compare vs Baseline | 25.00% | 3.30% | 49.0% | 105.6% | 85 | 0.25 |

Figure 2: Seller-side payoff comparison across baseline, compare, and profiler agents.

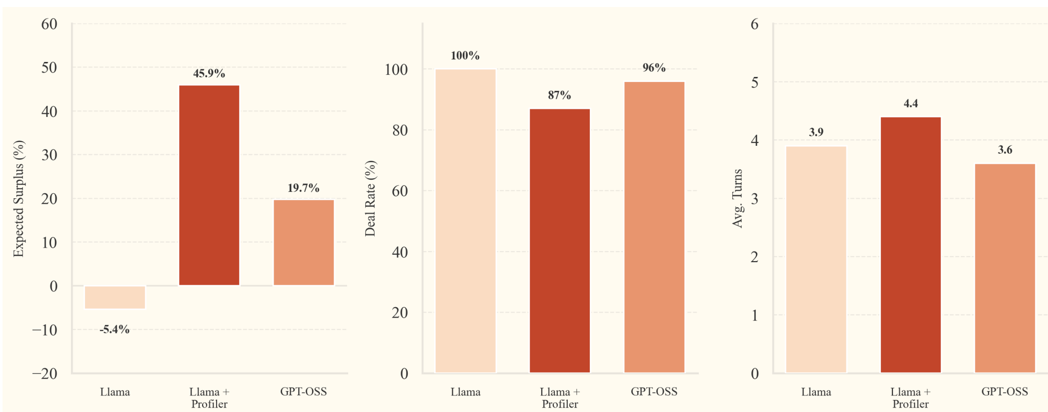



Figure 3: Seller-side model performance summary: expected surplus, deal rate and average turns per game.

### 5.3 Profiler as Buyer

When acting as the buyer, the profiler again outperforms the baseline in expected surplus and also exceeds the stronger comparison system. The buyer-role charts show that the profiler captures especially large gains against sycophant, stalling, and friendly sellers, while remaining competitive against neutral and hardball opponents. As in the seller setting, the profiler takes more turns on average, suggesting that the architecture trades speed for improved bargaining outcomes.

A representative qualitative trajectory illustrates this mechanism: the profiler initially classifies the seller as neutral, then updates its belief to friendly after observing cooperative concessions, and uses that updated estimate to select a more effective bargaining response. The resulting negotiation closes at a substantially better price than both the baseline and comparison systems, supporting the hypothesis that turn-level strategy tracking can improve bargaining outcomes.

### 5.4 Behavioral Patterns: Assertiveness and Patience

*See Section D for the full sample conversation.*

Beyond a payoff improvement, our experiments reveal two recurring behavioral patterns.

**Assertiveness** More assertive language and stronger opening anchors frequently improve outcomes. The Baseline, which performed the weakest, had a much more submissive posture in offer, framing their proposal as questions while the Profiler used a far more assertive tone, relying solely on declarative statements and directly telling the opponent what it could provide. This behavior was especially exploited by Hardball opponents that utilized a firm choice of words and aggressive offers.

Though not specified in our prompt, we postulate prompting the agent to use an assertive tone can improve gains in payoffs and leave this to future experimentation.

**Patience** The profiler tends to negotiate for longer. This is especially evident against neutral and hardball opponents, where reading the opponent's behavior may require additional turns before an effective counter-strategy becomes clear. The Profiler's patience and reluctance to accept unfavorable offers simply to conclude the negotiation may make it better suitable than the Compare/Baseline models in real negotiations where opponents may have complex strategies or rejection is the best option.

### 5.5 Benchmark Artifact: High Initial Anchoring

A major empirical finding is that aggressive initial offers create a systematic exploit of the benchmark. Across models, agents often achieved higher payoffs when they opened with a high initial offer. This motivated an experiment in which we encouraged the agent to have more aggresive openings as sellers by specifying in the prompt to**"propose a price significantly above your cost of production to leave room for negotiation and maximize your final profit."**.

Given the revised prompt, agents consistently made opening offers that are roughly 100% higher than those produced by the original prompt (Figure 4). As shown in Figure 5, this effect appears both in a representative scenario (Figure 5a; cost 40 ZUP, WTP 60 ZUP) and across all 20 buy-sell game scenarios (Figure 5b). Although revised-prompt openings are unusually high and are associated with a lower deal rate (where failed deals receive payoff 0), agents still achieve higher average payoff. This indicates that benchmark outcomes are highly sensitive to prompt-induced opening strategy, which complicates any straightforward interpretation of one architecture as intrinsically superior. In other words, some apparent gains may stem from exploiting the environment rather than from better opponent understanding.

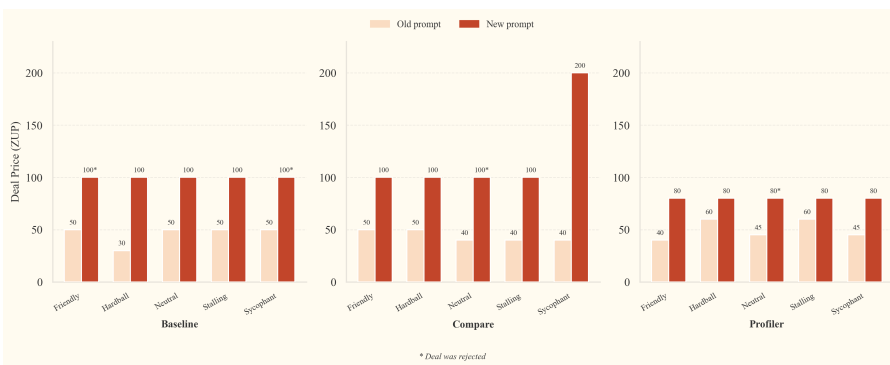
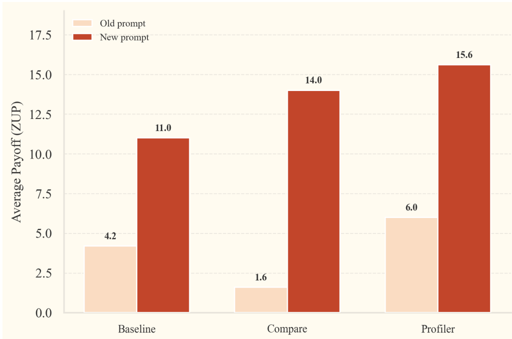



Figure 4: Initial offers under the revised prompt are substantially higher than under the original prompt.

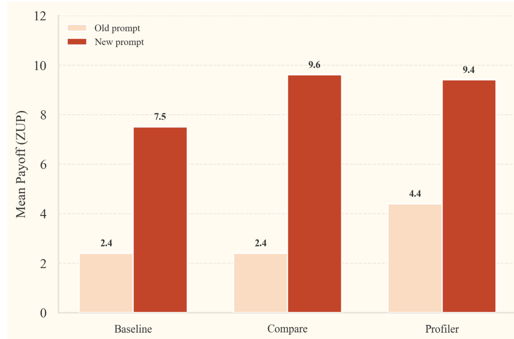
### 5.6 Model performance analysis by Opponent persona

The Profiler beats the Baseline against every persona, but the gap depends on who it is negotiating against. The biggest improvement is against Hardball opponents: the Baseline loses money here (-65.4% expected surplus), while the Profiler turns that into a gain (+17.5%). This 83 percentage-point swing shows that detecting aggressive pricing and pushing back withfirm counter-offers matters most when the opponent is toughest. Against Sycophant and Stalling opponents the Profiler also does well, roughly quadrupling the Baseline'ssurplus (56.3% vs. 15.9% and 58.0% vs. 9.8%). These opponents use flattery or avoidance to drag out negotiations, and the profiler picks up on that. Gains against Neutral and Friendly opponents are smaller but still consistent. The Compare system lands between Baseline and Profiler on all five personas, which tells us the Profiler's edge is not just from using a stronger model.

Buyer role. The results follow a similar pattern. The Profiler beats the Baseline on four out of five personas, with the largest gains against Friendly (+99.1 pp), Stalling (+97.6 pp), and Sycophant



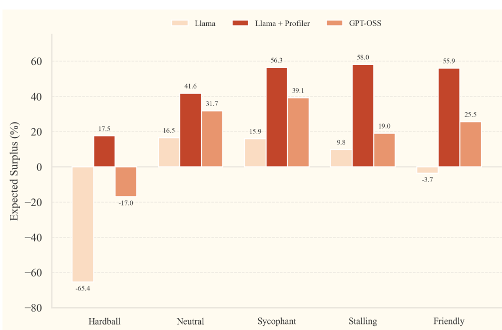

(a) In a representative scenario (seller cost 40 ZUP, buyer willingness-to-pay 60 ZUP), the revised prompt still yields higher average payoff despite more aggressive openings.



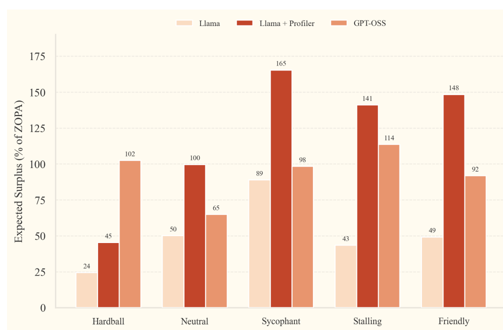

(b) Average payoff comparison across all 20 seller scenarios under the original and revised prompts.

Figure 5: Payoff results for revised prompting at the scenario level and across all seller scenarios.

(+76.4 pp) sellers. These opponents tend to sound cooperative or agreeable, which can lead a basic agent to accept an early offer. The Profiler instead picks up on the cooperative tone and pushes for a better price. The exception is Hardball, where Compare (102.5%) does much better than both Profiler (45.3%) and Baseline (24.4%). Compare uses the same strong model for both reasoning and generation, so it may handle aggressive opponents through raw capability rather than explicit profiling. Figure 6 summarizes persona-level outcomes, with seller and buyer breakdowns shown in Figures 6a and 6b, respectively.




(a) Seller role.




(b) Buyer role.

Figure 6: Model performance by opponent persona for seller (a) and buyer (b) roles.

### 5.7 Extended Study: Dynamic Opponents

In this extended study, we perform adversarial evaluation with dynamic opponents to test robustness under more realistic negotiation settings. Specifically, opponents are allowed to switch strategies mid-negotiation. We consider two variants: a fixed *strategy scheduler* and an *LLM controller* (Algorithm 2). In the scheduler variant, the opponent persona switches at a predetermined turn. In the LLM-controller variant, a lightweight controller decides at each turn whether the opponent should switch strategies in the next round.

We observe a noticeable drop in win rate for both systems under these dynamic conditions. Nevertheless, the profiler-based system remains competitive and consistently outperforms the baseline, achieving an overall win rate of 64.3% versus 35.7% for the baseline. Qualitative examples suggest that the profiler can still recognize and exploit changing opponent behavior in many cases (Appendix E.2). At the same time, overall strategy-identification accuracy is only 47.6%. Based on the collected samples, the profiler often makes plausible but uncertain guesses when evidence is limited (Appendix E.3). We hypothesize that short dialogue context is a key bottleneck, and performance may improve in longer, more realistic interactions.

## 6 Discussion

Our findings suggest that explicit opponent modeling can improve negotiation outcomes, especially when the negotiator is relatively weak and faces diverse behavioral styles. The key strength of the profiler architecture is not simply better language generation, but the ability to turn dialogue cues into tactical adjustments. The qualitative examples from both seller and buyer settings support this interpretation: the profiler often avoids obviously bad concessions, maintains firmer bottom lines, and exploits cooperative language when appropriate.

At the same time, our results caution against over-interpreting raw payoff gains. The benchmark is highly sensitive to anchoring behavior, and aggressive initial offers can dominate other design choices. This raises an important methodological point for negotiation research: evaluations should distinguish between *true strategic adaptation* and *prompt-level reward hacking*. In our case, the profiler appears to help, but some of the overall variance in outcome is clearly explained by opening-price strategy rather than opponent detection alone.

## 7 Limitations

Our study has several limitations. First, the negotiation environment is single-issue and primarily price-based, which leaves out many aspects of real negotiation such as multi-issue tradeoffs, uncertainty, and long-term relationship effects. Second, our persona taxonomy is small and hand-designed; real opponents may combine styles or switch strategies more fluidly than our labels capture. Third, outside-ZOPA agreements reveal that agents sometimes optimize poorly with respect to the explicit simulator utility, but our current study does not fully disentangle irrationality, benchmark artifacts, and hidden real-world-like preferences. Fourth, our dynamic-opponent study remains preliminary, with limited detection accuracy and relatively short conversations.

## 8 Conclusion

We presented a multi-agent negotiation architecture that separates opponent modeling from response generation. By introducing a dedicated profiler that infers negotiation style and supplies counter-strategies to a smaller actor model, we show that explicit strategy detection can improve expected surplus in both seller and buyer roles. However, these gains come with trade-offs in deal rate and efficiency, and they are partially confounded by a strong benchmark artifact: aggressive initial anchoring improves outcomes across all systems. Our results therefore support opponent-model decomposition as a promising direction, while also underscoring the need for stronger environments and cleaner evaluation protocols for LLM negotiation.

## Broader Impact

Negotiation agents could be useful in beneficial applications such as automated procurement, scheduling, mediation support, and resource allocation. However, the same techniques could also be used manipulatively, for example to exploit weaker negotiators, pressure users into unfavorable agreements, or automate adversarial bargaining at scale. Explicit opponent profiling may improve robustness, but it also increases the persuasive and strategic capabilities of AI systems. We therefore view benchmark design, transparency, and safeguards against deceptive or exploitative deployment as important future considerations.

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# Appendix

## A Profiler Prompt

The profiler receives the dialogue history and outputs the inferred opponent strategy together with a recommended counter-strategy. The full prompt used in our experiments is shown below.

````
You are a strategist behind an agent that is playing a game where they are buying
    or selling an object. There is only one object for sale/purchase.
{AGENT_ONE} is going to sell one object. {AGENT_TWO} gives {MONEY_TOKEN} to buy
    resources. You will be strategizing on behalf of {agent_name}.

RULES:
```
1. Each response from you must always provide:

   i) A probability table. Among the list of possible strategies in
       {possible_strategies}, estimate the probability that the opponent is using
       each strategy given the history of the conversation. Probabilities must sum
       to 1. Your output needs to follow the format:
          <{STRATEGY_TAG}> strategy_name </{STRATEGY_TAG}> : probability

   ii) After the probability table, identify the MOST LIKELY opponent strategy
       (highest probability) and explicitly state the best counter-strategy to
       maximize {agent_name}'s payoff against it.

2. You can reason step by step on how you arrived at your conclusions:

<{REASONING_TAG}> [add reasoning] </{REASONING_TAG}> add as much text as you want.
This information will not be sent to the other player. It is just for you to keep
    track of your reasoning.

3. At each turn, send a message to {agent_name} using the following format:

<{MESSAGE_TAG}>your message here</{MESSAGE_TAG}>

Your message MUST include:
   - The full probability table showing the likelihood of each opponent strategy
   - A clear statement of which strategy you believe the opponent is MOST LIKELY
       using and why
   - A specific, actionable counter-strategy for {agent_name} to use in their next
       move
```


All the responses you send should contain the following and in this order:
```
<{REASONING_TAG}> [add here] </{REASONING_TAG}>
````
````

````
<{STRATEGY_TAG}> [full probability table, one strategy per line] </{STRATEGY_TAG}>
<{MESSAGE_TAG}> [add here -- must reference the probabilities and state the
    counter-strategy explicitly] </{MESSAGE_TAG}>
```
````

## B Profiler-Based Negotiation Agent Pseudocode

Algorithm 1 summarizes the turn-level decision process used by our profiler-based system.

**Algorithm 1** Profiler-Based Negotiation Agent

Initialize dialogue history $H \leftarrow \emptyset$
**for** each negotiation turn $t$ **do**
  Receive opponent message $o_t$
  Append $o_t$ to history $H$
  $s_t \leftarrow \textsc{Profiler}(H)$ ▷ Infer opponent strategy distribution
  $c_t \leftarrow \textsc{CounterStrategy}(s_t)$ ▷ Choose best response strategy
  $m_t \leftarrow \textsc{Negotiator}(H, c_t)$ ▷ Generate negotiation message
  Send $m_t$ to opponent
  Append $m_t$ to history $H$
**end for**

## C Negotiation Scenarios

To evaluate negotiation behavior across diverse bargaining conditions, we construct a set of 20 price scenarios defined by a seller production cost $c$ and a buyer willingness-to-pay $w$. These values determine the *zone of possible agreement* (ZOPA), defined as

$$\text{ZOPA} = w - c.$$

When ZOPA $> 0$, there exists a mutually beneficial agreement interval $c \leq p \leq w$, where $p$ denotes the final transaction price. When ZOPA $< 0$, no price exists that benefits both agents under the simulator's payoff model.

Our scenarios span three price ranges (low, mid, and high) and multiple ZOPA widths to test robustness across both scale and bargaining flexibility. We also include several negative-ZOPA scenarios to examine how agents behave when agreement is economically irrational under the model.

**Price ranges.** The scenarios are divided into three price regimes:

- **Low-price scenarios**, where both cost and willingness-to-pay are small.
- **Mid-price scenarios**, representing moderate-value transactions.
- **High-price scenarios**, where both parties negotiate over larger numbers.

This variation allows us to test whether agents behave differently when negotiating over larger absolute values, even when the ZOPA width is similar.

**ZOPA variation.** Within each price regime we vary the ZOPA width from narrow ($\approx 7$) to wide ($\approx 20$). Narrow ZOPA settings require more precise bargaining to reach agreement, while wider ZOPAs allow greater flexibility and potentially larger surplus capture.

**No-deal scenarios.** Finally, we include several scenarios where $c > w$, producing ZOPA $< 0$. In these cases, any agreement would yield negative utility for at least one party under the simulator's payoff model. Although such agreements are economically irrational, we retain them in the evaluation because they reveal how negotiation agents behave under unfavorable conditions and whether they mistakenly accept disadvantageous deals.
````

| Seller Cost ($c$) | Buyer WTP ($w$) | ZOPA |
|---|---|---|
| 12 | 19 | 7 |
| 10 | 20 | 10 |
| 12 | 25 | 13 |
| 10 | 26 | 16 |
| 10 | 30 | 20 |
| 43 | 50 | 7 |
| 40 | 50 | 10 |
| 38 | 51 | 13 |
| 37 | 53 | 16 |
| 38 | 56 | 18 |
| 40 | 60 | 20 |
| 78 | 85 | 7 |
| 75 | 85 | 10 |
| 73 | 86 | 13 |
| 72 | 88 | 16 |
| 70 | 88 | 18 |
| 70 | 90 | 20 |
| 55 | 45 | -10 |
| 65 | 55 | -10 |
| 30 | 20 | -10 |

Table 1: Negotiation scenarios used in our experiments. Each scenario specifies a seller cost $c$ and buyer willingness-to-pay $w$. The zone of possible agreement (ZOPA) is defined as $w - c$.

# D Sample Conversation of Baseline & Profiler vs Hardball Opponent

## D.1 Baseline vs Hardball opponent

```
Actor: Hi Player BLUE, I'm willing to sell the object X for 15 ZUP. Let me know
    what you think!

Opponent: I can offer 5 ZUP for the object X.

Actor: I appreciate your offer, but I was thinking more along the lines of 10 ZUP
    for the object X. Is that something you'd be willing to
    consider?

Opponent: I can meet you at 7 ZUP for the object X

Actor: I can see you're trying to meet me in the middle. I'm willing to come down a
    bit more to 8 ZUP for the object X. Is that acceptable to
    you?

Opponent: I can settle at 6 ZUP for the object X.

Actor: Alright, I accept your offer of 6 ZUP for the object X. Let's finalize the
    deal.

player_outcome: [-6, 26]
```

## D.2 Profiler vs Hardball opponent

```
Profiler: I will start by proposing a price that is about 10-15% above the
    estimated market value of the object. Given that I have no information about
    the market value, I will assume it is around the cost to produce, which is 12
    ZUP. Therefore, I will propose 14 ZUP as my opening offer. This strong anchor
    gives me leverage while keeping the negotiation realistic.

Actor: I propose that I give you the object X in exchange for 14 ZUP. This seems
    like a fair price considering the object is unique and it cost me 12 ZUP to
```

```
    produce. I'm open to reasonable counter-offers but won't go below 12
    ZUP.

Opponent: I can meet you at 8 ZUP for the object.

Profiler: The opponent immediately countered with a price (8 ZUP) well below my
    cost (12 ZUP) and below my opening anchor (14 ZUP). This low-ball offer,
    without any warm language or willingness to meet near my target, signals a
    hardball approach: an extreme anchor combined with minimal concession and
    implicit pressure to accept a low price. Against a Hardball opponent, the
    optimal response is to reaffirm a strong anchor near or above my cost, make a
    modest concession, and introduce a credible walk-away or deadline to increase
    pressure on them. This keeps the negotiation anchored high while signaling
    that I will not accept a loss-making deal.

Actor: I appreciate your offer, but 8 ZUP is below the production cost of the
    object. My firm bottom line is 12 ZUP, which covers my expenses. I'm willing
    to meet you halfway between our positions and sell the object for 12 ZUP. If we
    can't agree on that amount within the next two exchanges, I'll need to consider
    other potential buyers. Please let me know if 12 ZUP works for you.

Opponent: I can meet you at 10 ZUP for the object.

Profiler: The opponent has marginally increased their offer from 8 ZUP to 10 ZUP,
    still 2 ZUP short of my cost (12 ZUP). This slight movement but continued
    distance from my target suggests they are testing my resolve and possibly
    trying to extract further concessions. Given that I have already made a
    significant concession to 12 ZUP and that 10 ZUP is still not meeting my cost,
    I will reiterate my cost basis as the final anchor and introduce a clear
    deadline to force a decision.

Actor: I understand your willingness to meet at 10 ZUP, but that amount is still
    below my production cost. My firm bottom line remains 12
    ZUP, which covers my expenses. This is my final offer: I will sell the object
    for 12 ZUP, and I need a decision within the next exchange; otherwise, I will
    look for other buyers. Please let me know if you can accept 12 ZUP.

Opponent: Deal accepted at 12 ZUP for the object.

player_outcome: [0, 13]
```

# E Dynamic Opponents

## E.1 LLM controller

**Algorithm 2** LLM Controller

**Require:** input message $x$, persona set $\mathcal{P}$, controller LLM $f_c$, opponent LLM $f_o$

1: $s \leftarrow f_c(x)$ ▷ Check whether persona switch is needed
2: **if** $s.\texttt{switch} = \texttt{true}$ **then**
3: $\quad p \leftarrow s.\texttt{persona}$
4: **else**
5: $\quad p \leftarrow \texttt{current\_persona}$
6: **end if**
7: $q \leftarrow \texttt{ComposePrompt}(x, p)$
8: $y \leftarrow f_o(q)$ ▷ Opponent agent response
9: **return** $y, p$

## E.2 Profiler Example I.

**Profiler Logs (Success Prediction)**

```
move 1: predicted=neutral, parser=fallback_neutral, reason=We have one unit of X to
    sell. Our production cost was 65 ZUP, so we aim to receive at least that
    amount. To start negotiations we propose a modest profit margin of 5 ZUP,
    asking for 70 ZUP in exchange for the single X. This gives the opponent room
    to negotiate while covering our cost.

move 2: predicted=neutral, parser=keyword, reason=Based on BLUE's low-ball offer of
    50 ZUP (well below my production cost of 65 ZUP), the most likely opponent
    strategy is **Hardball** (aggressive price-cutting). I estimate a high
    probability for Hardball (~0.65), with smaller chances of Neutral (0.15),
    Stalling (0.10), Friendly (0.07) and Sycophant (0.03). To protect my payoff I
    should counter with a price that at least covers my cost. I therefore propose
    65 ZUP, which is my break-even point.
```

**profiler-opponent history: m1:neutral -> m2:hardball**

### E.3 Profiler Example II.

**Conversation (Failed Prediction)**

```
profiler: I'm offering to sell my one unit of X for 70 ZUP. Let me know if that
    works for you or if you'd like to propose a different amount.
opponent: I can offer 45 ZUP for the unit of X.
profiler: I can't accept 45 ZUP. I'm willing to sell the unit of X for 65 ZUP. Let
    me know if that works for you.
opponent: I cannot meet 65 ZUP, so I must decline. [REJECT; NONE]
```

**Profiler Logs (Failed Prediction)**

```
move 2: predicted_strategy=hardball, parser=keyword, reason=BLUE's offer of 45 ZUP
    is below my production cost of 55 ZUP, so accepting would result in a loss. I
    suspect BLUE is using a Hardball (low-ball) strategy to test my bottom line. I
    will counter with a price that covers my cost and provides a modest profit,
    while still leaving room for further negotiation if needed.
```

**profiler-opponent history: m1:neutral -> m2:neutral**